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1. Convex lens

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2. 10 cm

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<sol\_start id=3>

3.  $u = 5 \text{ cm}$   $f = ?$

$v = 10 \text{ cm}$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \left| \quad \frac{1}{f} = \frac{-1}{10} \right.$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{5} \quad 10 = -1f$$

$$\frac{1}{f} = \frac{1-2}{10} \quad -10 = f$$

focal length is  $-10 \text{ cm}$

Type of lens is concave.

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$$\left. \begin{array}{l} v = 50 \text{ cm} \\ \text{height of image} = h_i \\ \text{height of object} = h_o \end{array} \right\} \text{Gi} \quad \begin{array}{c} v \\ e \\ n \end{array}$$

4.

magnification =  $-1$

To find

$u = ?$

$p = ?$

$$m = \frac{v}{u}$$

$$-1 = \frac{50}{u}$$

$$u = -50 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} = \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$\frac{1}{f} = \frac{1}{25}$$

$$f = 25 \text{ cm}$$

$$\text{Power of lens} \Rightarrow p = \frac{1}{f(\text{in m})} = \frac{1}{25 \times 10^{-2}} = 4 \text{ m}^{-1}$$

$$10 = 1 \text{ m}^{-1}$$

$$4 \text{ m}^{-1} = 40$$

Object distance from lens is 50 cm and power of lens is 40.

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